| Date | 1/10/2026 | Maximum Marks | 4 Marks |
|---|---|---|---|
| Team ID | SWTID-2026-4326 | | |
| Project Name | Streamlining IT Procurement automating laptop orders with flow designer | | |

# Project Design Phase-II

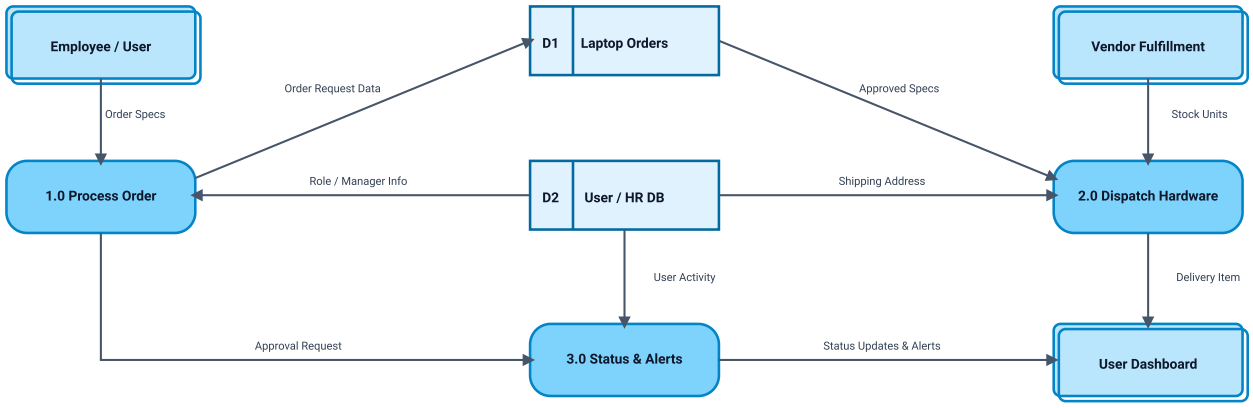
## Data Flow Diagram & User Stories

**Data Flow Diagrams:**
A Data Flow Diagram (DFD) is a traditional visual representation of information flows within a system. A clear DFD depicts system requirements graphically—showing how data enters and leaves the system, what changes the information, and where data is stored.

### User's View Data Flow (Simplified Sequence):

1. **User Input:** Employee logs into ServiceNow Self-Service Portal and submits laptop specifications and delivery address.
2. **Automated Trigger:** System sends request data to Flow Designer engine for rule-based budget and role verification.
3. **Approval & Routing:** Automated approval routing notifies Manager; upon sign-off, procurement order data is stored in Orders DB.
4. **Fulfillment & Dispatch:** Procurement system passes dispatch details to Hardware Vendor / Fulfillment center.
5. **Tracking & Confirmation:** Vendor pushes live tracking status to User UI; user confirms laptop receipt.

**Example: DFD Level 0 (Industry Standard — User's View)**



## User Stories

Comprehensive functional requirements breakdown capturing both **User Perspectives** (end-user interaction & goals) and **Designer / System Views** (UI/UX specs, Flow Designer logic, & integration criteria).

| User Type | Functional Req. (Epic) | Story ID | User Story / Task (User & Designer Views) | Acceptance Criteria | Priority | Release |
|---|---|---|---|---|---|---|
| **Customer** (Employee - Portal User) | Hardware Selection | **USN-1** | **[User View]:** As an employee, I can browse laptop configurations so that I can select a model suitable for my role. **[Designer View]:** Design role-filtered catalog cards displaying clear hardware specs and visual badges. | Display filtered laptops based on user department; compare up to 3 models side-by-side. | High | Sprint-1 |
| | Order Submission | **USN-2** | **[User View]:** As a user, I want to submit a laptop order with auto-filled employee details so that I save time. **[Designer View]:** Bind ServiceNow input form to Workday API to auto-populate Cost Center and Manager ID. | Zero manual entry for personal info; form validation triggers before Flow submission. | High | Sprint-1 |
| | Status Tracking | **USN-3** | **[User View]:** As a user, I can track my laptop order approval and delivery status in real-time. **[Designer View]:** Create a 4-step visual progress bar UI connected to Flow Designer stage updates. | Real-time stage badge updates (Submitted → Approved → Dispatched → Delivered). | Medium | Sprint-2 |
| **Approving Manager** | 1-Click Approval | **USN-4** | **[User View]:** As a manager, I can approve or reject laptop requests directly from my email/Teams. **[Designer View]:** Build Actionable Email Cards and Teams Bot webhooks powered by Flow Designer. | Manager can click "Approve" inside Teams without opening the full portal page. | High | Sprint-1 |
| | SLA Escalation | **USN-5** | **[User View]:** As a manager, I receive automated reminders so I don't delay team onboarding. **[Designer View]:** Configure Flow Timer trigger to auto-escalate pending orders after 24 hours. | Auto-escalation email sent to Director if manager does not respond within SLA window. | Medium | Sprint-2 |
| **IT Administrator & Designer** | Flow Designer Rules | **USN-6** | **[User View]:** As an IT Admin, standard laptop requests are auto-approved to speed up delivery. **[Designer View]:** Implement automated conditional logic in Flow Designer checking budget & stock. | Orders under budget threshold bypass manual approval and trigger vendor API. | High | Sprint-1 |
| | Vendor Integration | **USN-7** | **[User View]:** As an Admin, hardware orders are dispatched to suppliers automatically. **[Designer View]:** Build REST API integration in Flow Designer to transmit PO JSON to vendor API. | Automated PO creation and receipt of courier tracking number within 15 minutes. | Medium | Sprint-2 |
| **Customer Care Executive** | Support & Feedback | **USN-8** | **[User View]:** As a support agent, I can view order history to assist users with setup queries. **[Designer View]:** Consolidate user ticket history and hardware serial numbers into a single agent portal view. | Support agent can view full audit trail of order approvals and courier status. | Low | Sprint-2 |